

BOGD, Mongolia

WMO: 443380

Lat: 44.6742N

Lon: 102.1756E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-27.1	-33.7	0.2	-26.7	-31.2	0.3	-25.0	12.9	-6.3	11.6	-6.6	0.4	160	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	31.3	16.4	29.8	15.5	28.3	14.8	18.7	27.3	17.5	26.3	16.4	25.3	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	14.7	22.9	14.4	13.4	21.5	13.2	12.4	20.7	63.1	27.6	58.5	26.6	54.8	25.4	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	10.2	9.3	DB	-32.8	34.3	3.2	1.8	-35.0	35.6	-36.9	36.6	-38.7	37.7	-41.0	39.0
			WB	-32.9	19.2	3.1	2.5	-35.2	20.9	-37.0	22.4	-38.7	23.7	-40.9	25.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-16.3	-12.0	-2.6	7.0	13.2	18.7	21.6	19.7	13.5	4.8	-5.1	-12.7	(6)
(7)		DBStd	14.00	6.16	6.50	6.61	5.53	5.02	3.93	3.17	3.37	4.56	5.23	5.67	4.91	(7)
(8)		HDD10.0	3321	816	616	391	120	27	1	0	0	20	172	454	704	(8)
(9)		HDD18.3	5392	1074	849	649	341	169	41	7	24	154	419	703	963	(9)
(10)		CDD10.0	1214	0	0	1	29	127	262	360	301	124	12	0	0	(10)
(11)		CDD18.3	246	0	0	0	1	11	52	108	66	8	0	0	0	(11)
(12)		CDH23.3	2489	0	0	0	3	112	573	1102	637	62	0	0	0	(12)
(13)		CDH26.7	723	0	0	0	0	17	162	368	171	4	0	0	0	(13)
(14)	Wind	WSAvg	3.5	2.5	3.0	3.8	4.4	4.8	3.6	3.5	3.2	3.6	3.5	3.5	2.8	(14)
(15)	Precipitation	PrecAvg	150	2	3	6	9	13	25	31	27	17	7	4	4	(15)
(16)		PrecMax	287	10	7	23	35	49	78	82	90	78	22	17	17	(16)
(17)		PrecMin	75	0	0	0	1	1	0	4	0	0	0	0	0	(17)
(18)		PrecStd	57	2	2	5	7	11	19	21	25	22	6	4	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	5.1	14.9	23.1	28.6	32.6	34.6	32.5	27.2	19.8	11.1	2.3	(19)
(20)		MCWB	-3.8	-0.6	4.2	10.4	11.2	17.2	19.3	16.6	13.0	8.9	3.4	-2.7	-2.7	(20)
(21)		2%	DB	-1.6	2.6	12.3	20.7	26.3	30.3	31.7	30.2	25.1	17.8	8.5	-0.2	(21)
(22)		MCWB	-5.9	-2.3	3.2	8.1	11.3	15.9	16.9	15.7	12.3	8.1	1.7	-4.5	-4.5	(22)
(23)		5%	DB	-4.2	0.5	9.9	18.7	24.2	28.5	30.2	28.6	23.4	15.4	6.0	-2.5	(23)
(24)		MCWB	-7.7	-3.9	1.8	7.4	10.4	14.5	16.0	15.0	11.2	6.3	0.0	-5.9	-5.9	(24)
(25)		10%	DB	-6.6	-1.7	7.6	16.5	22.1	26.7	28.7	26.9	21.4	13.4	3.9	-4.6	(25)
(26)		MCWB	-9.3	-5.3	0.5	6.1	9.6	14.0	15.5	14.4	10.5	5.0	-1.3	-7.5	-7.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.7	0.0	5.4	12.1	14.5	19.3	21.6	19.1	14.8	10.5	3.8	-2.5	(27)
(28)		MCDWB	0.7	4.5	12.8	21.6	24.9	30.4	31.5	26.8	24.0	18.6	10.4	2.0	2.0	(28)
(29)		2%	WB	-5.8	-2.1	3.5	9.6	12.6	17.7	19.4	17.9	13.6	8.6	2.0	-4.4	(29)
(30)		MCDWB	-1.7	2.4	11.7	18.8	22.9	27.4	28.3	25.9	21.7	16.2	8.1	-0.7	-0.7	(30)
(31)		5%	WB	-7.6	-3.9	2.0	7.8	11.5	16.4	17.9	16.8	12.5	6.8	0.2	-5.9	(31)
(32)		MCDWB	-4.3	0.3	9.4	17.2	21.9	25.8	26.6	25.4	20.3	14.7	5.7	-2.6	-2.6	(32)
(33)		10%	WB	-9.4	-5.3	0.6	6.5	10.4	15.2	16.7	15.8	11.4	5.5	-1.2	-7.5	(33)
(34)		MCDWB	-6.7	-1.8	7.3	15.7	20.7	24.4	25.5	24.2	19.8	12.7	3.6	-4.7	-4.7	(34)
(35)	Mean Daily Temperature Range	MDBR	11.8	12.3	12.7	11.9	11.9	11.7	11.2	11.4	11.4	11.5	10.8	11.2	(35)	
(36)		5% DB	MCDBR	13.3	13.2	13.7	13.5	13.6	13.7	13.3	13.3	12.5	12.6	11.8	12.2	(36)
(37)		MCWBR	10.1	9.7	8.5	7.4	6.8	6.3	5.2	5.4	5.8	7.2	7.7	9.4	(37)	
(38)		5% WB	MCDBR	13.1	12.9	13.4	13.2	12.8	12.7	11.6	11.0	11.6	12.3	11.2	12.0	(38)
(39)		MCWBR	10.1	9.6	8.8	7.9	7.2	6.8	5.7	5.0	6.0	7.7	7.6	9.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.225	0.250	0.286	0.321	0.337	0.350	0.343	0.327	0.298	0.273	0.240	0.225	(40)	
(41)		taud	2.430	2.374	2.303	2.257	2.261	2.301	2.362	2.391	2.435	2.444	2.452	2.423	(41)	
(42)		Ebn at Noon	930	955	952	940	930	916	918	924	930	914	899	892	(42)	
(43)		Edh at Noon	69	89	111	128	132	127	118	111	97	83	67	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.09	3.20	4.77	6.14	7.13	7.10	6.71	6.12	5.19	3.78	2.43	1.79	(44)	
(45)		RadStd	0.17	0.26	0.21	0.26	0.34	0.39	0.31	0.28	0.20	0.14	0.18	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-2.59	N/A	-1.25	-1.41	N/A	N/A	N/A	N/A

Nomenclature: See separate page